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SDC205 Lab

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Graphing Dynamically Generated Data

SDC205L

sdc205l 5.3 project - dynamically generated data.py - C:\PythonFiles\sdc205l 5.3 project - dynamically...

File Edit Format Run Options Window Help

```python
from datetime import datetime
from openpyxl import Workbook
from openpyxl.chart import BarChart, LineChart, Reference
import pandas as pd
import matplotlib.pyplot as plt

# This function writes comma-separated data to a CSV file.
def insertData(path, data):
    try:
        with open(path, 'a') as file:
            file.write(data + "\n")
        return True
    except:
        print("Error: Unable to write to file.")
        return False

# This function reads and displays the contents of a CSV file.
def viewData(path):
    try:
        print(f"Reading from file: {path}")
        with open(path, 'r') as file:
            for line in file:
                print(line.strip())
    except:
        print("Error: Unable to read file.")

# This function converts Fahrenheit to Celsius.
def convertData(data):
    converted_value = (data - 32) * 5 / 9
    return converted_value

# This function gets user input, converts the value, and saves the data to a CSV file.
def getInput():
    path = "C:\\PythonFiles\\ZooData.csv"
    entries = int(input("How many entries are being entered? "))

    for i in range(entries):
        entry_date = input("Enter the date: ")
        value = float(input("Enter the temperature in Fahrenheit: "))

        # convertData(data) requires a numerical value and returns the converted Celsiu
        converted_value = convertData(value)

        data = f"{entry_date},{value},{converted_value:.2f}"

        try:
            if insertData(path, data):
                print(f"The following data was saved at {datetime.now()}: {data}.")
        except:
            print("Error: Unable to save data.")

# This function asks the user for a graph type and generates a report chart from the CS
```

Ln: 83 Col: 0

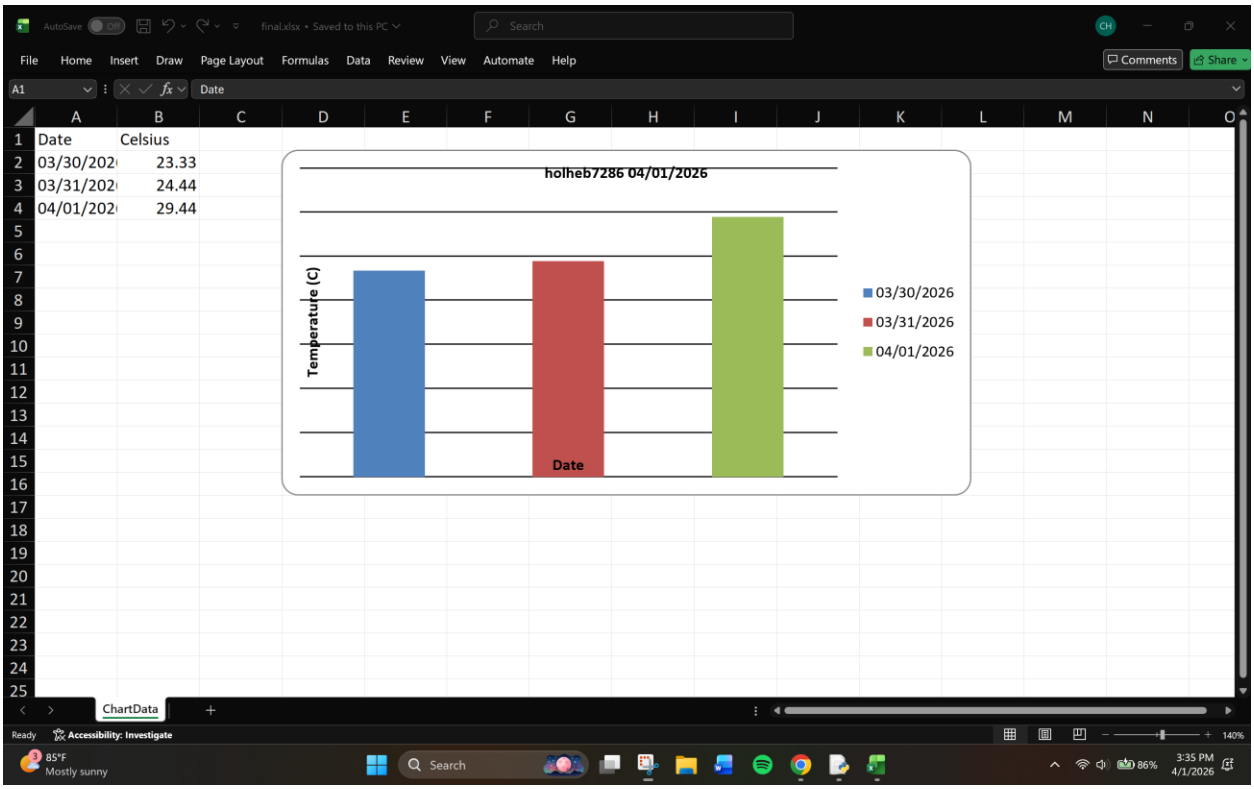
IDLE Shell 3.13.7

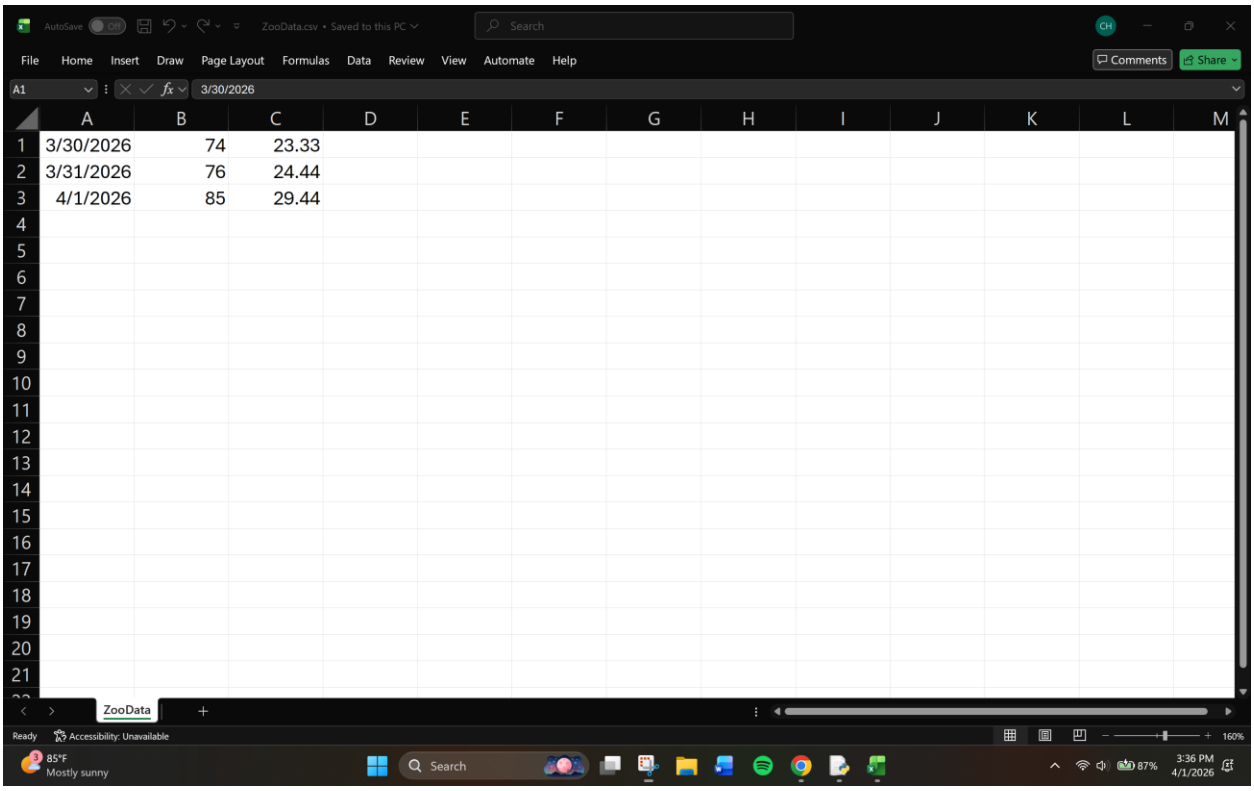
File Edit Shell Debug Options Window Help

```
Python 3.13.7 (tags/v3.13.7:bcee1c3, Aug 14 2025, 14:15:11) [MSC v.1944 64 bit (AMD
64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
== RESTART: C:\PythonFiles\sdc205l 5.3 project - dynamically generated data.py =
Enter your Student ID: holheb7286
holheb7286's Spreadsheet Automation Menu
1. Input Data
2. View Current Data
3. Generate Report
Select a menu option (1-3): 1
You selected menu option 1 at 2026-04-01 15:29:19.563095
How many entries are being entered? 3
Enter the date: 03/30/2026
Enter the temperature in Fahrenheit: 74
The following data was saved at 2026-04-01 15:29:32.477645: 03/30/2026,74.0,23.33.
Enter the date: 03/31/2026
Enter the temperature in Fahrenheit: 76
The following data was saved at 2026-04-01 15:29:40.931092: 03/31/2026,76.0,24.44.
Enter the date: 04/01/2026
Enter the temperature in Fahrenheit: 85
The following data was saved at 2026-04-01 15:29:50.781964: 04/01/2026,85.0,29.44.
>>>
== RESTART: C:\PythonFiles\sdc205l 5.3 project - dynamically generated data.py =
Enter your Student ID: holheb7286
holheb7286's Spreadsheet Automation Menu
1. Input Data
2. View Current Data
3. Generate Report
Select a menu option (1-3): 2
You selected menu option 2 at 2026-04-01 15:30:02.581292
Reading from file: C:\PythonFiles\ZooData.csv
03/30/2026,74.0,23.33
03/31/2026,76.0,24.44
04/01/2026,85.0,29.44
>>>
== RESTART: C:\PythonFiles\sdc205l 5.3 project - dynamically generated data.py =
Enter your Student ID: holheb7286
holheb7286's Spreadsheet Automation Menu
1. Input Data
2. View Current Data
3. Generate Report
Select a menu option (1-3): 3
You selected menu option 3 at 2026-04-01 15:30:16.789918
Enter chart type (line or bar): Bar
Choose data source: 1 for Fahrenheit, 2 for Celsius: 2
Chart created successfully in final.xlsx
>>>
```

Ln: 47 Col: 0

85°F Mostly sunny | Search | 86% | 3:35 PM 4/1/2026


final.xlsx • Saved to this PC
File Home Insert Draw Page Layout Formulas Data Review View Automate Help
Comments Share
A1 Date
Date Celsius
03/30/202 23.33
03/31/202 24.44
04/01/202 29.44
holheb7286 04/01/2026
Temperature (C)
Date
03/30/2026
03/31/2026
04/01/2026
ChartData
Ready Accessibility: Investigate
85°F Mostly sunny
3:35 PM 4/1/2026


ZooData.csv • Saved to this PC
File Home Insert Draw Page Layout Formulas Data Review View Automate Help
Comments Share
A1 3/30/2026
3/30/2026 74 23.33
3/31/2026 76 24.44
4/1/2026 85 29.44
ZooData
Ready Accessibility: Unavailable
85°F Mostly sunny
3:36 PM 4/1/2026